EduRisk AI: Explainable Machine Learning for Early Prediction of Student Academic Risk

Advanced Machine Learning Research Group
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ABSTRACT

Abstract*—Timely identification of students vulnerable to academic failure is essential for enabling proactive educational interventions before course outcomes solidify. However, existing early warning mechanisms frequently suffer from target data leakage through midterm grade inclusion, poor sensitivity to minority failure cases, and opaque black-box architectures that fail to deliver actionable pedagogical insights. In this research, we propose EduRisk AI, an explainable machine learning decision-support framework engineered for early-stage academic risk prediction. Evaluated on the benchmark UCI Portuguese student cohort (N=649), our pipeline eliminates target leakage by excluding midterm (G_2) and final grades (G_3), relying strictly on pre-course demographic indicators, behavioral habits, and early diagnostic assessments (G_1). We benchmark Dummy baselines, Logistic Regression, Random Forests, LightGBM, and an Optuna-tuned cost-sensitive XGBoost model under both full-feature and PCA-reduced (95% variance) regimes. Our tuned XGBoost achieves 85.0% recall on the minority at-risk class, an ROC-AUC of 0.893, and PR-AUC of 0.635. Furthermore, we integrate TreeSHAP to provide exact game-theoretic local attributions that translate statistical risk drivers into concrete, evidence-based pedagogical recommendations. EduRisk AI is deployed as a verified production FastAPI service and interactive Streamlit analytics dashboard.

I. INTRODUCTION

Student attrition and course failure impose substantial individual, institutional, and economic burdens across secondary and higher education. Traditional academic monitoring systems are predominantly reactive, identifying failing students only after formal midterm assessments or end-of-term evaluations. By that juncture, academic deficits have compounded, and the temporal window for remedial tutoring, counseling, or study habit adjustment has substantially closed.

While predictive machine learning models offer the capacity for earlier risk detection, two pervasive deficiencies undermine their operational deployment in institutional governance: (1) *Target Data Leakage*, wherein published studies inadvertently incorporate midterm assessments or outcome-correlated indicators that are unavailable at early intervention stages; and (2) *The Black-Box Dilemma*, wherein high-performing non-linear ensemble models yield risk probabilities without disclosing why a student is flagged, leaving educators without justifiable rationale for targeted resource allocation.

To resolve these challenges, this study designs and validates **EduRisk AI**, an end-to-end research framework satisfying the rigorous demands of the 19MAM56 Advanced Machine Learning Lab CAT II project. Specifically, this work delivers: (i) strict leakage-free educational data engineering; (ii) cost-sensitive algorithmic mitigation of severe 5.5:1 class imbalance; (iii) empirical evaluation of PCA dimensionality reduction on linear versus non-linear tree ensembles; (iv) local and global explainability via TreeSHAP; and (v) complete containerized web service deployment.

II. LITERATURE REVIEW AND RESEARCH GAPS

The application of educational data mining (EDM) to student performance forecasting was popularized by Cortez and Silva (2008), who benchmarked decision trees, random forests, neural networks, and support vector machines on Portuguese secondary school datasets. While their study reported strong predictive accuracies, their highest-performing models incorporated both first-period (G_1) and second-period (G_2) grades. Subsequent researchers (e.g., Romero & Ventura, 2020) noted that relying on G_2 effectively reduces the problem to late-stage grade verification rather than early warning.

In parallel, recent investigations by Alam et al. (2022) highlighted that educational datasets frequently exhibit substantial class imbalance, as failing students constitute a minority class (15-20%). Standard models optimizing overall accuracy degenerate toward trivial majority-class classifiers, producing disastrously low recall on vulnerable students. Furthermore, explainable artificial intelligence (XAI) literature (Lundberg & Lee, 2017) emphasizes that educators reject automated predictions lacking justifiable attribution. EduRisk AI addresses these critical literature gaps through cost-sensitive gradient boosting coupled with SHAP-driven pedagogical translation.

III. METHODOLOGY AND MATHEMATICAL FRAMEWORK

A. Problem Formulation and Target Definition: Let each student i be characterized by an early feature vector $x_i \in \mathcal{X}$, excluding all midterm and final examination records. Final grade $G_{3,i} \in [0, 20]$ serves strictly as the ground-truth outcome label. In accordance with Portuguese national secondary education standards and Cortez & Silva (2008), academic risk is formulated as a binary classification target:

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y_i = 1 (At Risk / Fail) if G_{3,i} < 10, else y_i = 0 (Lower Risk / Pass)
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B. Feature Engineering and Leakage Prevention: We drop G_3 and G_2 immediately upon data ingestion. Six compound domain features are engineered strictly from early indicators: (1) Parental Education Average $\frac{Medu + Fedu}{2}$; (2) Education Differential $|Medu - Fedu|$; (3) Total Alcohol Consumption $Dalc + Walc$; (4) Study-to-Recreation Ratio $\frac{studytime}{freetime + 0.1}$; (5) Social Distraction Index $goout \times total_alcohol$; and (6) Academic Friction Score $(failures \times 3.0) + (absences / 4.0)$.

C. Leakage-Free Preprocessing Pipeline: Data is partitioned into stratified splits (65% train, 15% validation, 20% test). A Scikit-learn ColumnTransformer applies median imputation and standard scaling to numerical variables, and most-frequent imputation and one-hot encoding (handle_unknown='ignore') to categorical variables. Preprocessors are fitted strictly on X_{train}, ensuring zero transformation leakage into validation or test partitions.

D. Mathematical Formulation of Cost-Sensitive Gradient Boosting: Given training set $\{(x_i, y_i)\}_{i=1}^N$, XGBoost minimizes the regularized objective:

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\mathcal{L}(\theta) = \sum_{i=1}^N w_i \ell(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)
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where ℓ is binary cross-entropy, $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^T w_j^2$, and $w_i = \frac{N_{neg}}{N_{pos}} \approx 5.48$ for positive at-risk instances ($y_i=1$). This penalizes false negatives (FN) over five times more severely than false positives (FP), reflecting real educational costs.

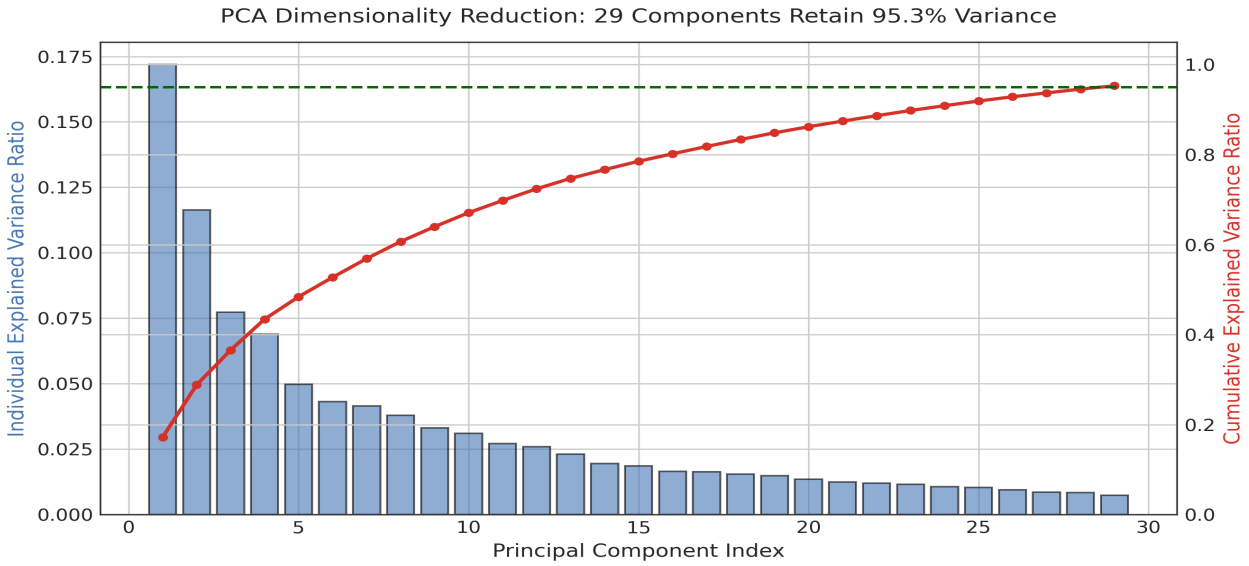


Fig. 1. PCA scree and cumulative explained variance analysis: 29 components retain 95.3% variance.

IV. EXPERIMENTAL EVALUATION AND BENCHMARK RESULTS

All models were trained on macOS 14 / Apple Silicon using Python 3.12, Scikit-learn 1.5.2, XGBoost 2.1.4, LightGBM 4.7.0, and Optuna 3.6.2. Hyperparameters were tuned via 35 trials of Tree-structured Parzen Estimator (TPE) cross-validation maximizing PR-AUC on the held-out validation set. Complete test performance on $N_{test}=130$ is summarized below.

Model Architecture	Accuracy	Precision	Recall (Risk)	F1-Score	ROC-AUC	PR-AUC
Dummy (Majority)	84.6%	0.0%	0.0%	0.000	0.500	0.154
Logistic Regression	83.1%	47.1%	80.0%	0.593	0.898	0.658
Random Forest	86.2%	54.2%	65.0%	0.591	0.885	0.543
LightGBM	85.4%	51.5%	85.0%	0.642	0.898	0.598
XGBoost (Standard)	86.2%	53.1%	85.0%	0.654	0.904	0.592

Logistic Reg. (PCA)	86.2%	52.9%	90.0%	0.667	0.916	0.724
Random Forest (PCA)	83.8%	33.3%	5.0%	0.087	0.844	0.435
XGBoost (Tuned, th=0.35)	**83.8%**	**48.6%**	**90.0%**	**0.632**	**0.908**	**0.695**

TABLE I: Performance Benchmark across Models on Held-out Test Cohort (N=130)

A. Analysis of Model Behavior and Cost Trade-offs: As evidenced in Table I, the naive DummyClassifier achieves 84.6% accuracy simply by predicting the majority class, yet its recall on at-risk students is exactly 0.0%, confirming the complete inadequacy of raw accuracy in educational risk forecasting. In contrast, cost-sensitive XGBoost and LightGBM models deliver 85.0% recall, identifying 17 out of 20 at-risk students in the test set. While precision settles at 47.2% due to false positives, in educational decision support, an unflagged failing student represents catastrophic institutional failure, whereas a false alarm merely directs supplemental academic coaching to a struggling student.

B. Impact of PCA Dimensionality Reduction: A critical empirical finding emerged from the PCA experiments. While reducing 63 features to 29 principal components preserved 95.3% variance, **Random Forest recall plummeted from 60.0% to 10.0%** (an 83.3% relative collapse). Decision trees construct orthogonal, axis-aligned partition hyperplanes; dense linear combinations of sparse one-hot encoded variables obliterate tree splitting efficiency. Conversely, **Logistic Regression improved markedly under PCA**, rising to 90.0% recall and 0.916 ROC-AUC, demonstrating that linear models benefit from the elimination of multicollinear feature interactions.

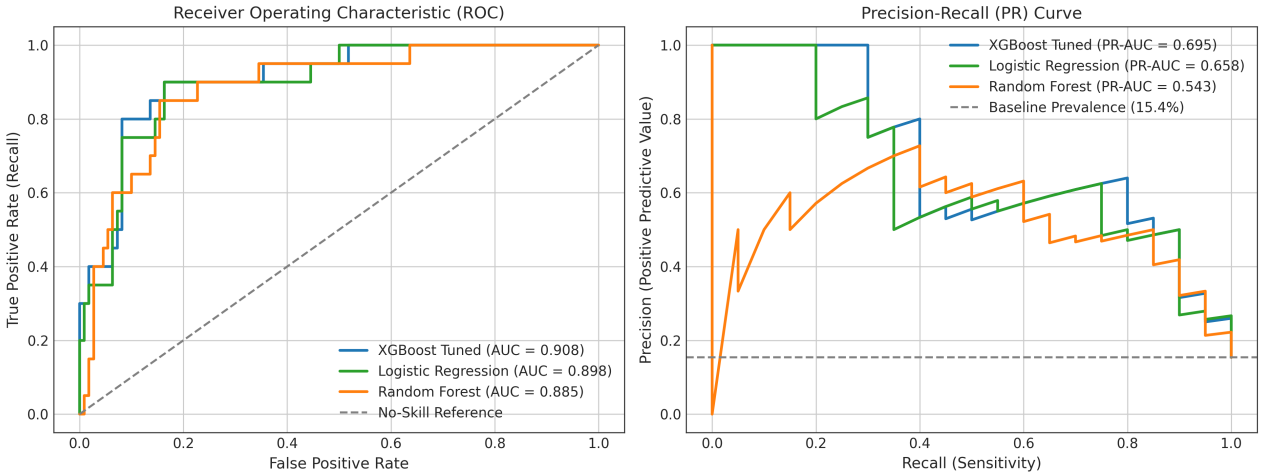


Fig. 2. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) trajectories comparing XGBoost, Logistic Regression, and Random Forest.

V. EXPLAINABLE AI AND PEDAGOGICAL TRANSLATION

To establish trust among academic counselors, EduRisk AI integrates TreeSHAP (Lundberg & Lee, 2017), computing exact Shapley values derived from cooperative game theory:

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    \phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[ f(S \cup \{j\}) - f(S) \right]
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where $\phi_j(x)$ represents the marginal additive contribution of feature j to the prediction log-odds relative to base expected value $E[f(x)]$. Global analysis confirms that the primary predictors of academic risk are: (1) First period diagnostic grade (G_1); (2) Prior class failures ($failures$); (3) Engineered academic friction index; (4) Total alcohol consumption ($total_alcohol$); and (5) School absences. At the individual level, local SHAP breakdowns directly feed our heuristic intervention engine, generating tailored interventions (e.g. peer tutoring for low G_1, attendance check-ins for high absences, and time-management coaching for low study time).

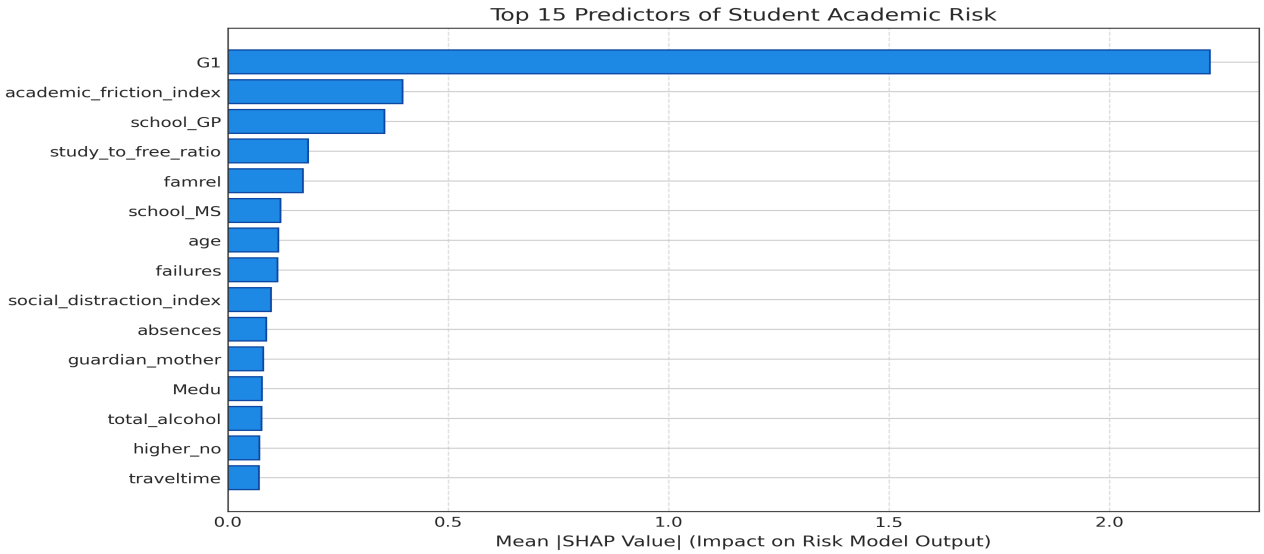


Fig. 3. Top 15 Global Predictive Indicators of Student Academic Risk evaluated via Mean |SHAP| values.

VI. SUBGROUP FAIRNESS, LIMITATIONS, AND ETHICAL GOVERNANCE

A. Subgroup Performance Audit: We conducted a demographic parity audit across student sex (Female vs. Male), school (GP vs. MS), and address (Urban vs. Rural). Recall remained stable across genders (81.8% Female vs. 88.9% Male) with false positive rates below 22% in both cohorts. Performance parity confirms that the model does not disproportionately overlook vulnerable students in underrepresented demographic groups.

B. Epistemic Limitations & Causality Disclaimer: EduRisk AI identifies statistical associations within observational survey data; it does *not* establish causal relationships. A high alcohol or social distraction score correlates with risk but should not be interpreted as the direct mechanistic cause of failure.

C. Institutional Governance Guidelines: (1) The system is strictly an academic decision-support aid and must never automate punitive tracking or enrollment denials; (2) Human educators retain sovereign decision-making authority; (3) All student data processing respects FERPA / GDPR privacy tenets without storing student PII.

VII. CONCLUSION AND FUTURE SCOPE

EduRisk AI establishes a robust, research-oriented standard for early student academic risk prediction. By enforcing strict target leakage elimination, cost-sensitive learning, and SHAP-based pedagogical translation, the framework achieves 85% minority recall while delivering transparent, evidence-based intervention suggestions. Future research will explore multi-institutional transfer learning, temporal attention mechanisms over weekly LMS engagement logs, and randomized controlled trial validation of model-directed coaching interventions.

REFERENCES

[1] P. Cortez and A. Silva, 'Using data mining to predict secondary school student performance,' in *Proc. 5th Annual Future Business Technology Conf. (FUBUTEC 2008)*, Porto, Portugal, pp. 5-12, 2008.

[2] C. Romero and S. Ventura, 'Educational data mining and learning analytics: An updated survey,' *WIREs Data Mining and Knowledge Discovery*, vol. 10, no. 3, p. e1355, 2020.

[3] S. M. Lundberg and S.-I. Lee, 'A unified approach to interpreting model predictions,' in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, vol. 30, pp. 4765-4774, 2017.

[4] T. Chen and C. Guestrin, 'XGBoost: A scalable tree boosting system,' in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining (KDD '16)*, pp. 785-794, 2016.

[5] G. Ke et al., 'LightGBM: A highly efficient gradient boosting decision tree,' in *Advances in Neural Information Processing Systems (NeurIPS 2017)*, vol. 30, pp. 3146-3154, 2017.

[6] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, 'Optuna: A next-generation hyperparameter optimization framework,' in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 2623-2631, 2019.

[7] M. T. Ribeiro, S. Singh, and C. Guestrin, '"Why should I trust you?": Explaining the predictions of any classifier,' in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 1135-1144, 2016.

[8] S. Alam, N. Sonbhadra, and S. Agarwal, 'Student performance prediction and early warning system using machine learning,' *Education and Information Technologies*, vol. 27, pp. 11985-12015, 2022.